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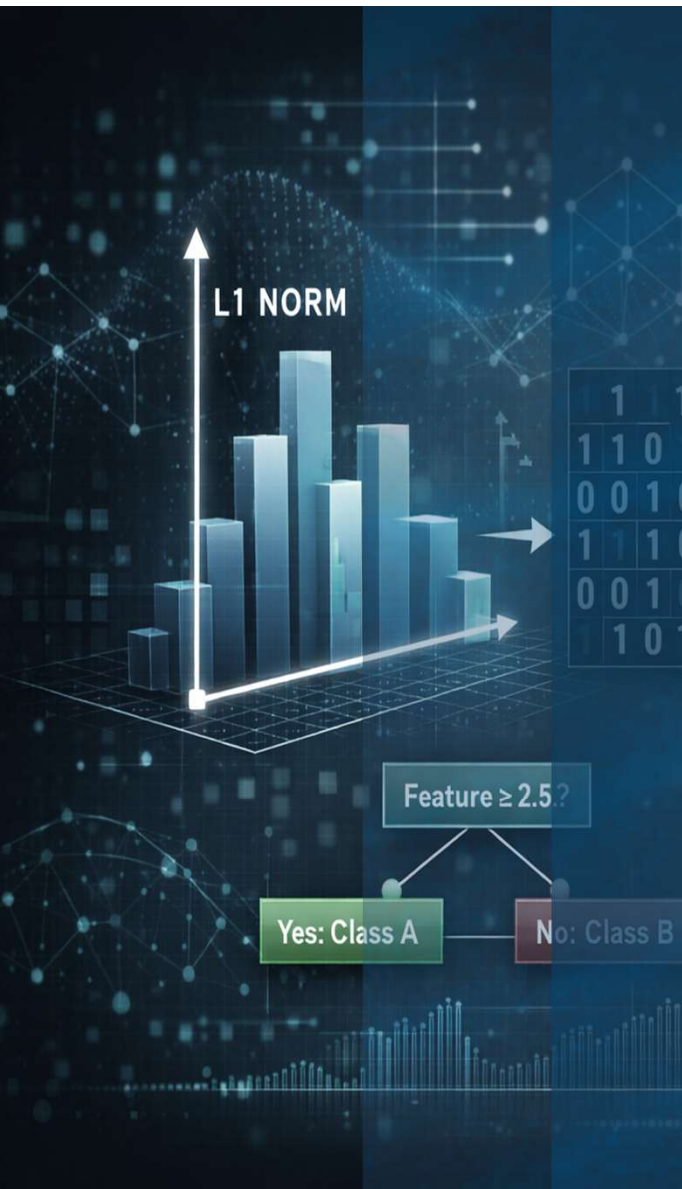


Interpretable Sparse Modeling of Longitudinal Signals via Critical-Range Rectification and Anytime Rule Compression

Achieving Explainability while Minimizing Computational Cost

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Overview



Thesis:

Critical-range rectification improves the reliability and interpretability of sparse longitudinal modeling; anytime rule compression makes the result operationally usable.

01. Problem and Pipeline

Introduce the core problem, knowledge gap, and pipeline.

02. Answers to Research Questions

RQ1: Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?

RQ2: Why does rectification work, and how far does the theoretical explanation generalize?

RQ3: Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?

03. Boundaries and Implications

Discuss limitations, scope boundaries, and practical implications.

04. Contributions and Future Work

Summarize the contributions, extensions, and closing implications.



01.

Problem and Pipeline

Introduce the core problem, knowledge gap, and pipeline.



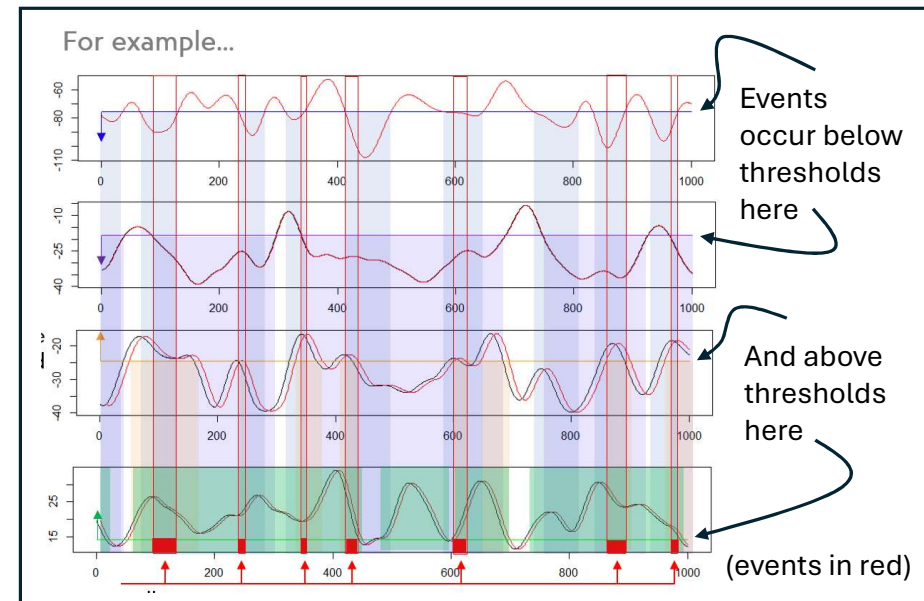
The Problem

➤ Three tensions:

- high dimension and correlation causing unstable support and weak lag attribution
- operational compute and model-complexity constraints
- high-stakes deployment requiring transparent and auditable rules

Real World

Longitudinal sensor, clinical, and industrial signals often reflect threshold behavior and lagged dependencies, complicating the creation of interpretable models.





Gaps in Prior Work

Accuracy, Interpretability, Sparsity, Efficiency

- Sparse penalties are mature but fragile under correlation and dependence.
- Structured/dependence-aware penalties often stay in raw correlated space.
- Rule learning is transparent but search-heavy with combinatorial scaling.
- Prior work lacked an end-to-end, reproducible synthesis across the method families reviewed.

Cross-Family Positioning

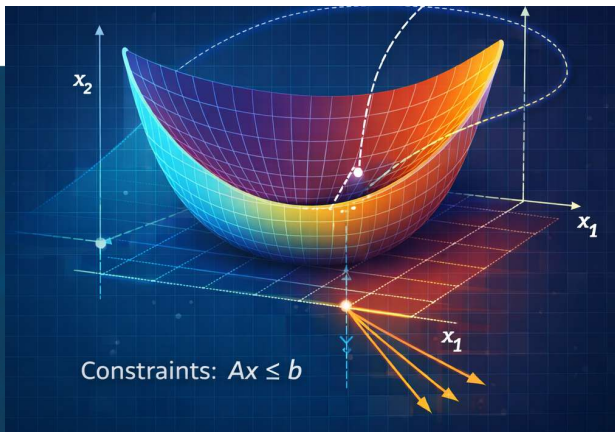
Several method families address parts of the problem identified in this dissertation, but each leaves a gap for interpretable, lag-aware, deployment-ready modeling.

L1 / elastic-net / adaptive lasso

Changes representation before fitting, rather than relying only on penalty tuning in the original dependent feature space.

Grouped / ordered sparsity

Targets threshold logic and lag semantics through feature redesign, not only coefficient constraints.



Interpretable additive / rule-ensemble models (EBM / RuleFit)

The dissertation pipeline produces a shared rectified sparse baseline and, when justified, a smaller policy-screened rule artifact.

Dependence-aware / Quadratic Programming / robust selectors

Couples redundancy reduction to explicit threshold semantics and a downstream rule path.

Direct rule-list / LAD-style methods

Restricts logic compression to active support rather than searching the full lag-expanded space.



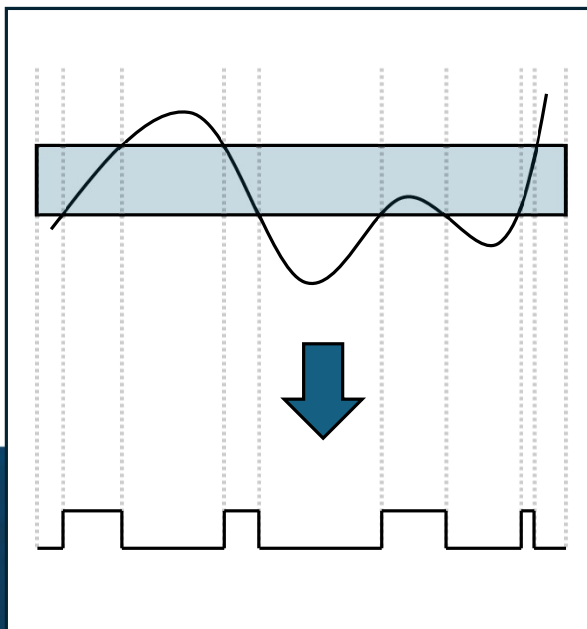
02.

Answers to Research Questions

The main thrust of the work.

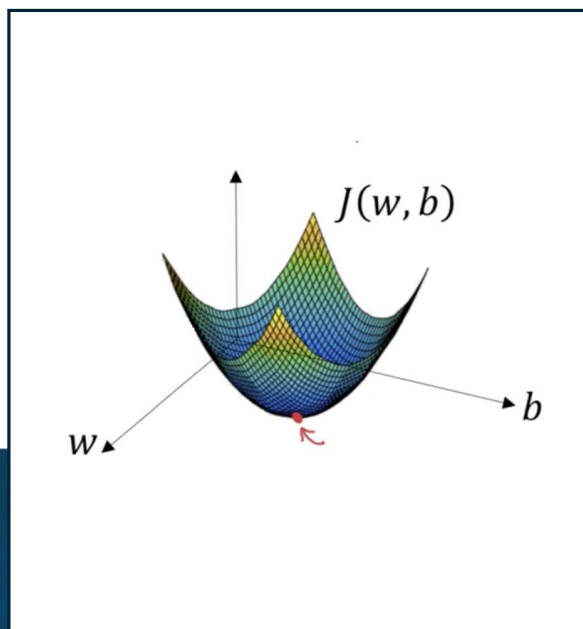


Final Pipeline (end-to-end)



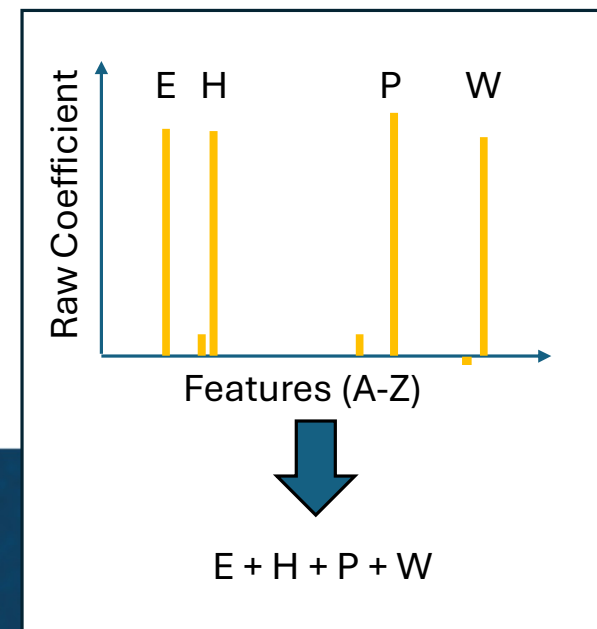
Rectify

Convert lagged continuous measurements into critical-range indicators learned from training data only.



Fit Sparse Model

Fit an L1-regularized logistic baseline on the rectified design to identify candidate feature-lag triggers.



Policy-controlled m-of-K Anytime Rule Compression

Compress active terms into an m-of-K rule when held-out deployment gates pass.

Critical-Range Rectification



Definition

A critical range is a train-only event-associated region of a feature's values. In the simplest independent-feature case, it is the minimum-to-maximum interval observed during event times. Rectification converts continuous, correlated signals into logical indicators.

-1

Not within the critical range

+1

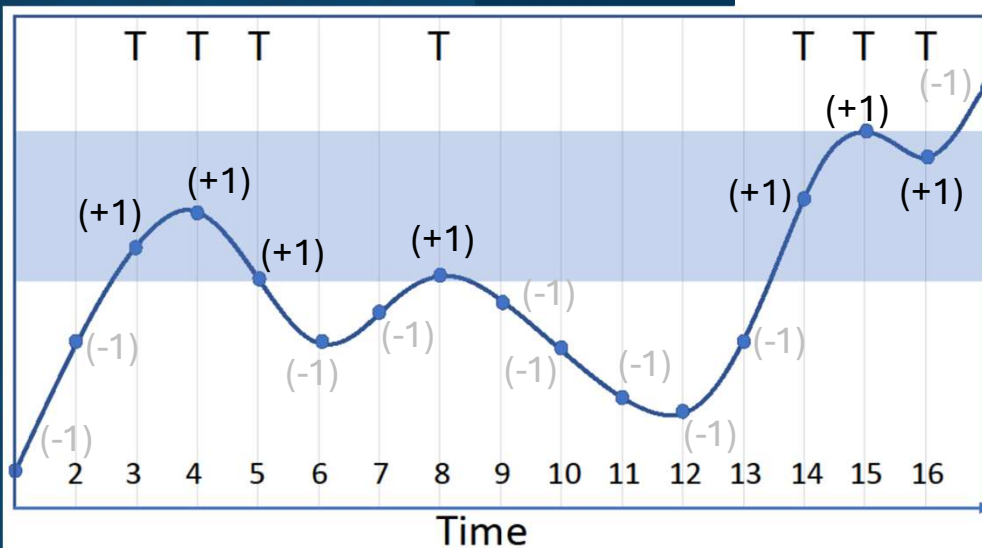
Within the critical range

Matrix

A rectified matrix with the same dimensions as the original is constructed, with $\{-1, +1\}$ replacing feature values.

Outcome

A response vector is constructed in which the event either occurs or does not occur $\{1, 0\}$



**** Ranges learned on training folds only**

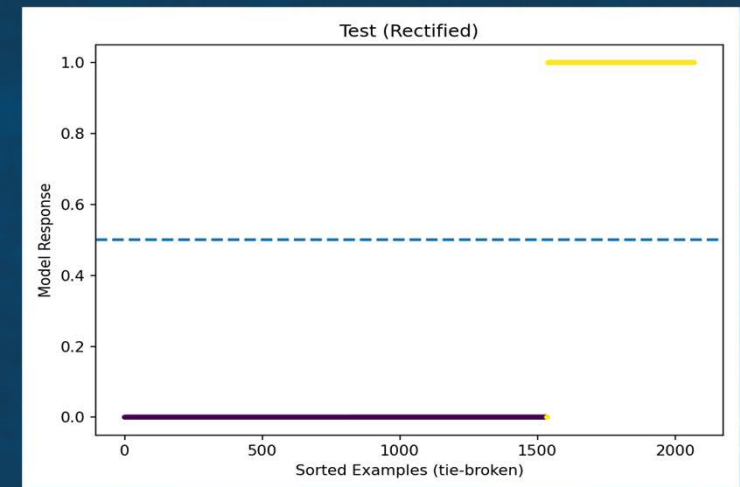
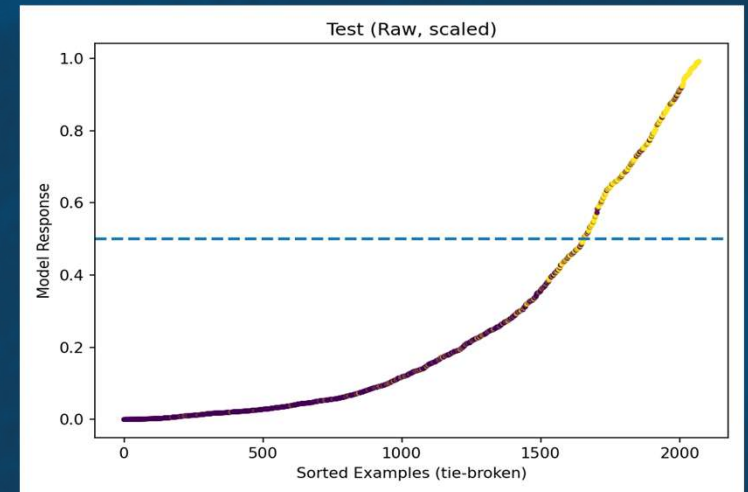


Measuring Success

Predictive and Interpretable

The goal is not only predictive accuracy, but reliable attribution, compact interpretation, and deployable rules with controlled performance loss.

- *Performance:*
 - *Area under the ROC curve (AUC)*
 - *Youden's J-index*
 - *Exact-lag F1*
- *Attribution quality and stability:*
 - *Active support size*
 - *Pairwise Jaccard / support stability*
 - *Feature-lag localization*
- *Compression / deployment quality:*
 - *Held-out non-inferiority gates*
 - *Rule length / number of rules*
- *Efficiency:*
 - *Training / inference runtime*



Research Questions

Motivation

Selecting relevant features in high-dimensional, strongly correlated longitudinal data is challenging when go/no-go thresholds and lags drive outcomes. Many physical, industrial, clinical, and behavioral systems are governed by such rules. Critical-range rectification maps continuous measurements to logical indicators defined by train-only critical ranges, simplifying correlations and stabilizing sparse model fitting. This motivation leads to the following research questions:

RQ1

Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?

RQ2

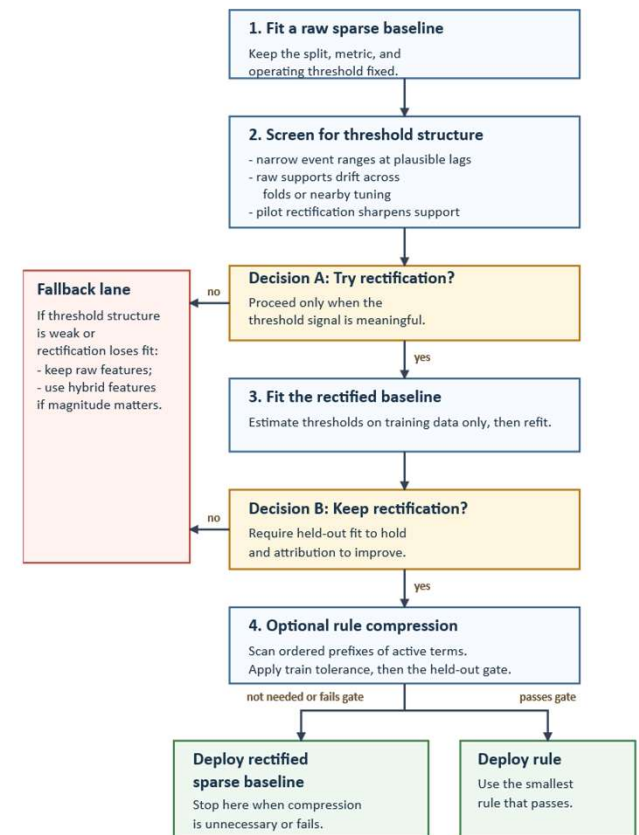
Why does rectification work, and how far does the theoretical explanation generalize?

RQ3

Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?



Applying the Pipeline to a New Dataset





02a.

Research Question #1

Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?

RQ1: Rectification Workflow



Collect and Flatten Longitudinal Data

Lagged examples are collected for each event, with each measurement offset across candidate time lags. This lag expansion creates highly correlated columns.



Estimate Critical Ranges

Using training data only, estimate event-associated minimum and maximum values for each feature-lag column.



Rectified L1 Logistic Fit

Fit an L1-regularized logistic model on the rectified design to determine the sparse coefficient set.

Algorithm 1 Binarization

Require: $\mathbf{X}, y$ (training set only)

Output: $\mathbf{A}, r_j[b, c]$

1: Let $r_j[b, c]$ be the critical range for feature vector $\mathbf{X}_j$, where b is the minimum and c is maximum defining that range.

2: **procedure** BINARIZATION($\mathbf{X}, y$)

3: Let Z be the rows of $\mathbf{X}$ where $y=1$.

$Z \subseteq \mathbf{X}$, where $Z = \{\mathbf{X} \mid y = \text{TRUE}\}$

4: Compute $r_{j=1..d} = [\min(Z_j), \max(Z_j)]$.

Note: If the min / max calculations induce brittleness in the solution (e.g. if there are outliers in the data set), using a robust quantile can provide a remedy.

5: Compute $\mathbf{A}$, the set of transformed features, such that:

$a_{ij} = +1$ when $x_{ij} \geq r_j[b]$ and $x_{ij} \leq r_j[c]$

$a_{ij} = -1$ when $x_{ij} < r_j[b]$ or $x_{ij} > r_j[c]$

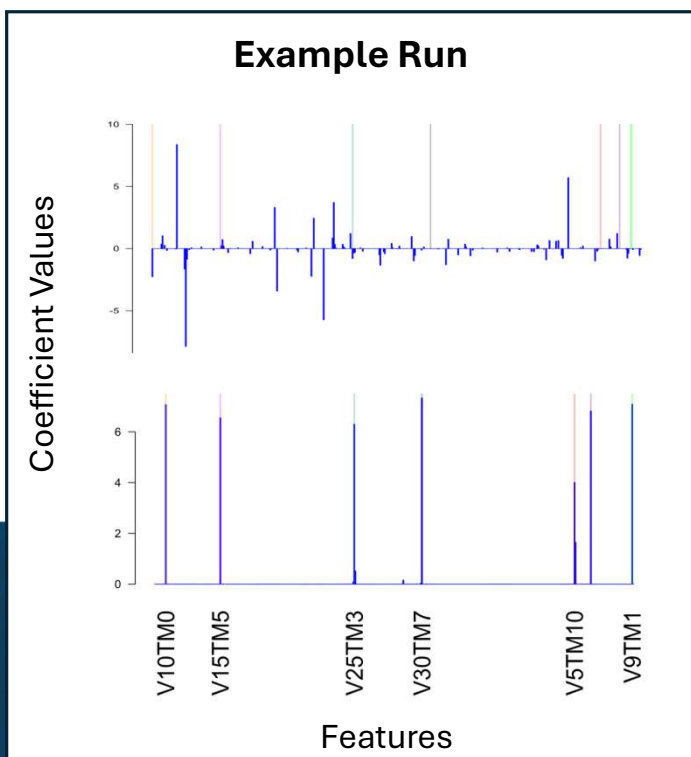
6: **end procedure**

Critical Range Determination

Measurements observed during event-positive training rows define the candidate critical range.



RQ1: Synthetic Data



Method	AUC	Youden's J-index	Exact-lag F1	Pairwise Jaccard	Total Nonzero	Runtime (s)
Raw	0.986 ± 0.008	0.882 ± 0.027	0.048 ± 0.082	0.056 ± 0.096	340.667 ± 105.396	23.142 ± 0.815
Rectified	0.998 ± 0.002	0.971 ± 0.019	0.619 ± 0.082	0.452 ± 0.090	190.000 ± 117.796	12.196 ± 1.562
Rule	0.994 ± 0.003	0.974 ± 0.015	0.759 ± 0.056	0.600 ± 0.346	6.000 ± 3.464	12.201 ± 1.562

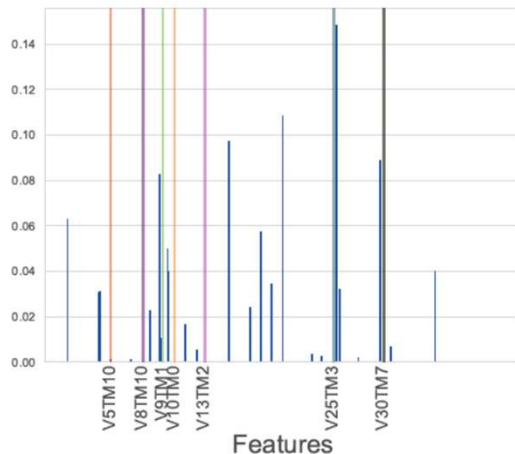
Compare Results across Raw vs. Rectified vs. Rule Sparse Fits

Across repeated synthetic resamples with known ground truth, rectification improved discrimination, support stability, exact-lag recovery, compactness, and runtime. The compressed rule preserved discrimination while reducing the active support to only a few terms.



RQ1: Quadratic Programming Comparison

Importance Magnitudes for Quadratic Programming
(Python qp-feature-selection, raw data)



- Katrutsa–Strijov QP provides feature-importance weights, but not a full predictive model for AUC, Youden's J, or F1 evaluation. For comparison, an L1 fit was applied to the selected feature set.
- Many importance values came close to identifying the correct variables, but in several cases selected an incorrect lag.

Method	AUC	Youden's J-index	Exact-lag F1
QP + L1	0.894	0.603	0.000
Rectified L1	0.997	0.946	0.571

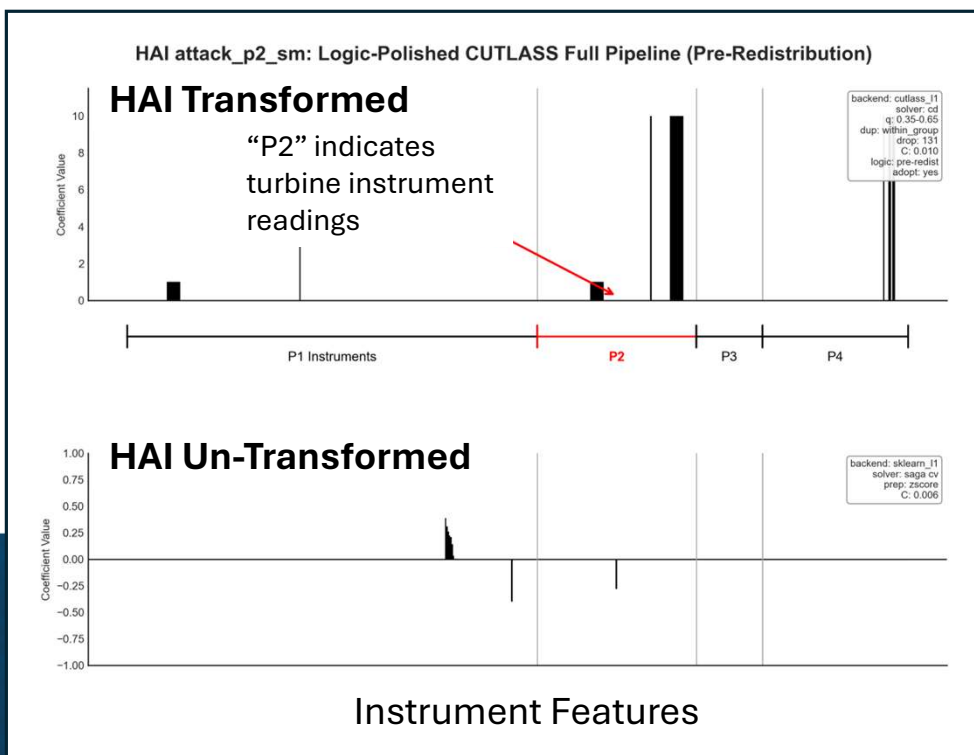
Both methods compared on a level playing field

The original Katrutsa–Strijov implementation was in R and ran much slower. For a level comparison, the method was reimplemented in Python; after reimplementation, runtime was comparable.



RQ1: Real World Data

HAI Exemplar and Cross-Domain Check



Statistic	HAI (Rectified)	HAI (Raw)	UNICEF** (Rectified)	UNICEF** (Raw)
ACC	0.987	0.939	0.902	0.963
AUC	0.982	0.943	0.865	0.989
F1	0.987	0.942	0.894	0.958
Youden's J-Index	0.974	0.878	0.815	0.925

** UNICEF scope note: raw metrics were higher, but the raw model used a less compact feature set (16 vs. 5 categories), illustrating that rectification is not uniformly superior and must be judged against attribution and complexity goals.

Rectified vs. Raw Feature Attribution

In the HAI exemplar, rectification concentrated coefficient mass on the turbine-loop P2 instruments while suppressing unrelated features. A smaller P4 response is plausible because P4 is the HIL simulator block and contains signals coupled to the P2 turbine-loop response.



RQ1 Answer

RQ1: “Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?”

The answer is **yes, conditionally**. Critical-range rectification improved support concentration, lag localization, compactness, and often runtime in threshold-and-lag aligned regimes. Cross-domain checks show that it should be used when threshold structure is meaningful and held-out performance is preserved.

Transition to RQ2:

- *What mechanism explains the empirical gains?*
- *Under what assumptions can rectification improve sparse recovery?*
- *Where do the theoretical and empirical boundaries appear?*

Efficiency Note

In the synthetic studies, rectification reduced runtime even after transformation overhead, while producing more compact and stable sparse supports.



02b.

Research Question #2

Why does rectification work, and how far does the theoretical explanation generalize?

RQ2: Irrepresentable Condition (IC)



LASSO support recovery is favored when the IC term is bounded:

$$\|X_S^{\top c} X_S (X_S^{\top} X_S)^{-1} \text{sign}(\beta_S)\|_{\infty} < 1$$

may become easier to satisfy after zero-threshold sign binarization under the theorem assumptions.

- Zhao and Yu's IC gives a formal lens for when LASSO can recover the true sparse support.
- Raw lag-expanded longitudinal features often violate this condition through high collinearity.
- RQ2 uses zero-threshold sign binarization as a tractable special case to show how rectification can contract dependence and improve IC behavior.
- The result is mechanistic and scoped, not a universal guarantee for every rectifier or dataset.

Scoped Theoretical Mechanism

The theorem analyzes zero-threshold sign binarization under explicit assumptions because that case admits a closed-form arcsin correlation relation. Practical critical-range rectification is broader, so later boundary checks test how far the mechanism appears to extend.



RQ2: The Arcsin Relation

The bivariate normal orthant probability identity, $P(X > 0, Y > 0) = \frac{1}{4} + \frac{1}{2\pi} \arcsin(\rho)$ implies the following closed-form mapping:

$$\tilde{\rho} = \frac{2}{\pi} \arcsin(\rho),$$

hence $|\tilde{\rho}| \leq |\rho|$ for $\rho \in (-1, 1)$

- For standardized, jointly normal variables with correlation ρ , zero-threshold sign binarization induces the closed-form arcsin mapping.
- The tilde (“~”) marks the binarized feature correlation.
- Under these assumptions, binarized pairwise correlations satisfy $|\tilde{\rho}| \leq |\rho|$.
- **Equality occurs only at $\rho \in \{-1, 0, 1\}$** ; otherwise, the transformed correlation is smaller in magnitude.

Key intuition

Sign binarization contracts pairwise correlations, reducing harmful dependence among lag-expanded features.



RQ2: Gram Matrix Bounds

The Gram matrix $X_S^T X_S$ is expressed as G and its inverse as G^{-1} in the following equicorrelated case assuming $\rho \in (-1/(s-1), 1)$ so that G is invertible:

$$G = n((1 - \rho)I_s + \rho J_s),$$

$$G^{-1} = \frac{1}{n(1 - \rho)} \left(I_s - \frac{\rho}{1 + \rho(s - 1)} J_s \right)$$

where, I_s is the identity matrix of size S , and J_s is a matrix of ones of size S .

- Similar expressions for the binarized versions $\tilde{G}$ and $\tilde{G}^{-1}$ and can be obtained by substituting $\tilde{\rho}$ for ρ .
- When the expression $0 < \tilde{\rho} < \rho < 1$ is true, it can be shown that:

$$\| \tilde{G}^{-1} \|_{\infty} \leq \| G^{-1} \|_{\infty}$$

- Thus, sign binarization contracts positive correlations in this equicorrelated setting, reducing the inverse-Gram contribution to the IC term. Recall that this is the second term in the IC:

$$\| X_S^T X_S (X_S^T X_S)^{-1} \text{sign}(\beta_S) \|_{\infty} < 1$$

Key bridge to IC

The inverse Gram matrix is part of the IC term. In the scoped equicorrelated case, sign binarization contracts correlations and can reduce the inverse-Gram contribution, making support recovery more favorable.



RQ2: Parsing the Results

Taking the original IC expression:

$$\|X_{S^c}^T X_S (X_S^T X_S)^{-1} \text{sign}(\beta_S)\|_\infty < 1$$

and defining a similar form:

$$\theta_j = C_{jS} G_{SS}^{-1} \text{sign}(\beta_S)$$

then applying the submultiplicative property of induced norms:

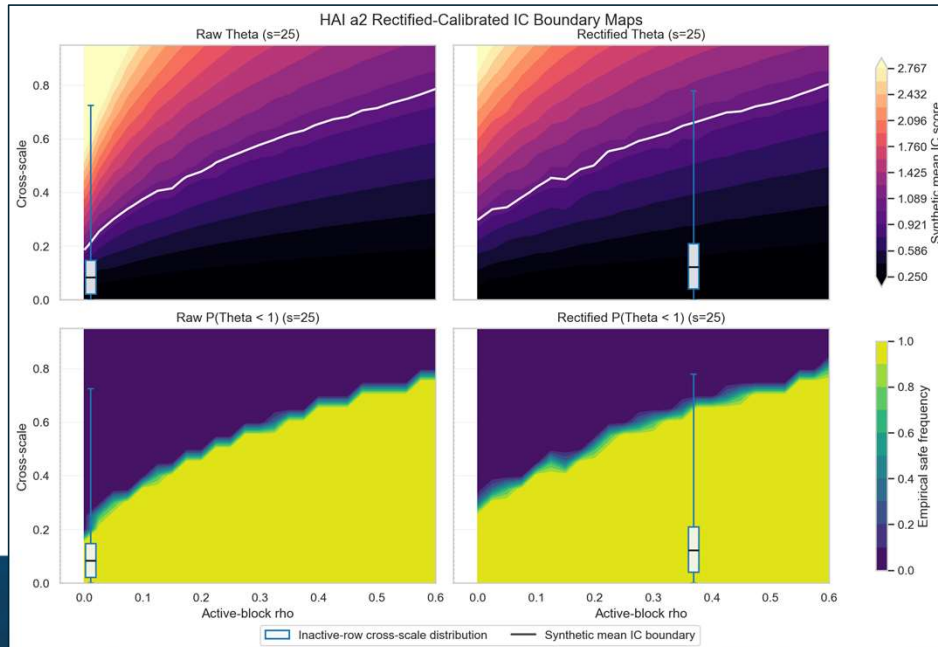
$$|\theta_j| \leq \|C_{jS}\|_\infty \|G_{SS}^{-1}\|_\infty \|\text{sign}(\beta_S)\|_\infty$$

- The sign vector $\text{sign}(\beta_S)$ has infinity norm 1 on the active support.
- The arcsin relation attenuates positive correlations after zero-threshold sign binarization.
- Under the scoped assumptions, inactive-active covariance terms $(X_{S^c}^T X_S)$ shrink.
- This reduces the ability of inactive lag features to imitate active lag features in the IC expression.
- So with these assumptions, $\|\tilde{C}_{jS}\|_\infty \leq \|C_{jS}\|_\infty$, often strictly smaller for $0 < |\rho| < 1$ (i.e. the inactive-active cross-correlation term is no larger after sign binarization under these assumptions).

Parsing the IC term

The IC term combines inactive-active covariance, inverse active-set covariance, and coefficient-sign direction. Under the scoped assumptions, sign binarization attenuates the covariance terms that let inactive features mimic active ones.

RQ2: Boundary Checks Beyond the Theorem



RQ2 takeaway

The theorem gives a scoped mechanism, not a universal guarantee. Rectification is strongest when the transformation contracts harmful dependence while preserving the event logic, and boundary checks can identify when caution or hybrid features are needed.

- Using the same IC-boundary framework as the theorem, I overlaid the HAI real-world data before and after rectification.
- Rectification shifts the empirical distribution into a more favorable region and increases contrast between active and inactive dependence structure.

Boundary Check	Empirical Result	Meaning
Positive zero-threshold sign case	Strong	Clean match to theorem assumptions
Shifted one-sided cutoff	Strong	Practical thresholds can preserve the mechanism
Shifted interval rule	Weak	Broader rectifiers need empirical validation
Negative-side IC sweep	Did not hold	Not all correlation structures benefit
Negative pair + complement feature	Recovered	Representation design can repair some failures

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Research Question #3

Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?

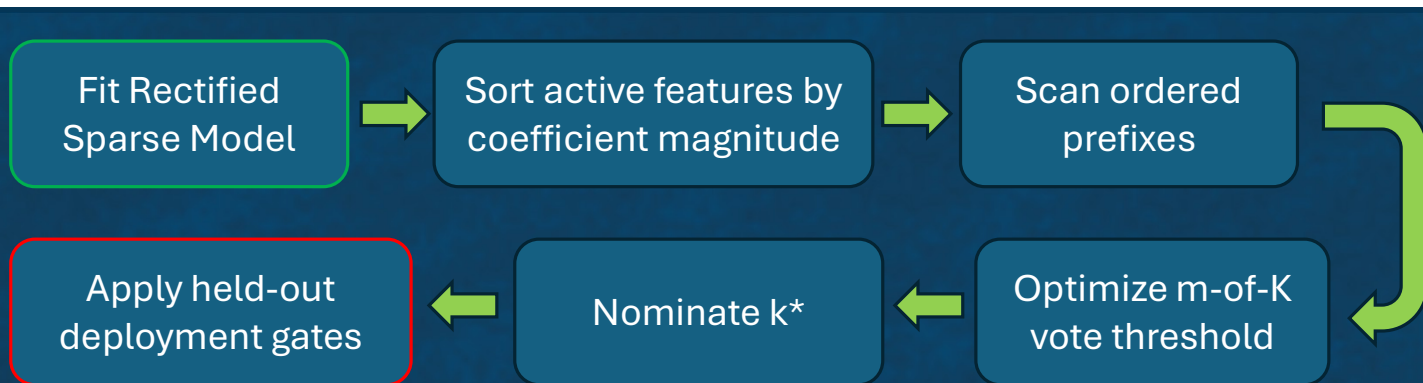
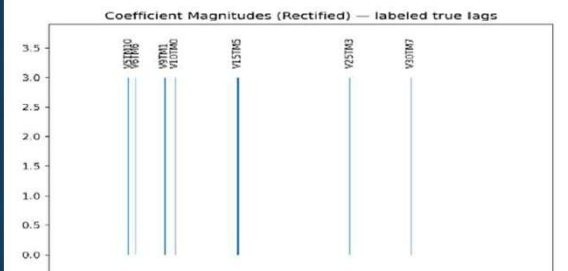
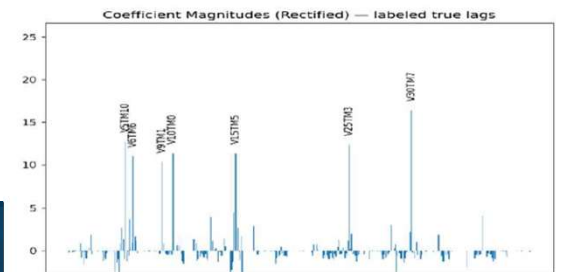
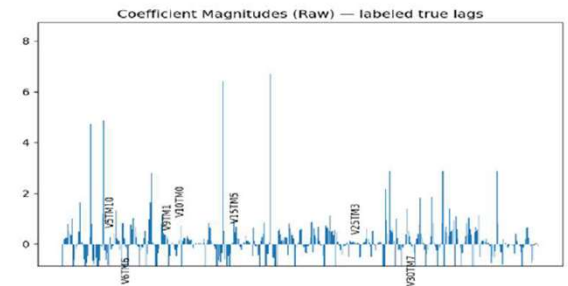
RQ3: Policy-Controlled Rule Compression

L1-regularized logistic models provide:

- Sparsity
- Computational simplicity, but...

Small residual coefficients *can obscure the underlying event logic*.

Rule compression converts the rectified sparse support into a compact m-of-K rule candidate, then accepts it only if held-out deployment gates pass.

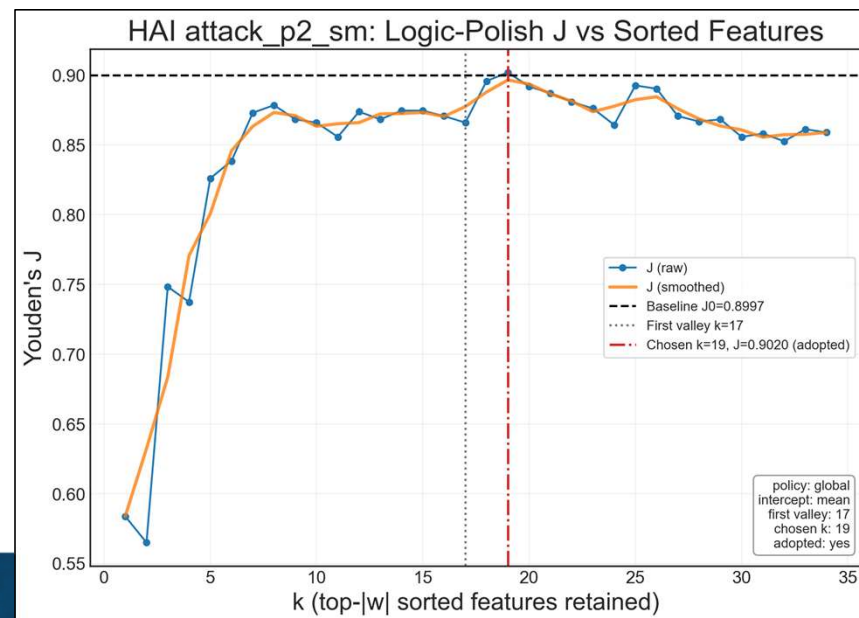


RQ3: HAI Compression Policy Results



The Youden's J vs K plot shows the number of features required to achieve each specific result for the HAI dataset.

- The anytime frontier scans ordered prefixes of active features and identifies compact m-of-K rule candidates.
- Compression is adopted only when the candidate passes held-out deployment gates; otherwise, the rectified sparse baseline is retained.
- In HAI, the policy selected compressed rules for P2 and P1/P2, but rejected candidates for P3 and P1/P3.



HAI Real World Data Test

Domain	Baseline nz	Candidate k*	Compression	ΔAUC	ΔJ	Deploy?
HAI Attack P2	39	14	2.79x	-0.011	+0.015	Yes
HAI Attack P1 & P2	25	8	3.12x	-0.001	-0.008	Yes
HAI Attack P3	23	7	3.29x	-0.003	+0.341	No*
HAI Attack P1 & P3	34	8	4.25x	+0.011	+0.014	No



03.

Boundaries and Implications

Discuss limitations, scope boundaries, and practical implications.

Scope Boundaries



Best-fit regime

- Threshold-mediated event structure
- Meaningful lagged dependencies
- Raw sparse models remain predictive but unstable
- Rectification improves attribution, compactness, or stability without harming held-out fit

Use caution when

- Threshold structure is weak or unstable
- Small threshold changes produce erratic results
- Post-threshold magnitude still carries important signal
- Compression fails held-out non-inferiority gates



Takeaway

Rectification is a representation strategy, not a universal replacement for raw continuous modeling.



Practical Implications

A Deployment Playbook

How to use the pipeline

1. Start with a raw sparse baseline.
2. Screen for threshold-and-lag structure.
3. Estimate critical ranges on training data only.
4. Compare raw vs. rectified attribution and held-out discrimination.
5. Compress only when a simpler rule is needed and justified.
6. Deploy compressed rules only when held-out gates pass.

What the user gets

- More stable feature-lag attribution.
- Smaller, more auditable sparse supports.
- Optional compact m-of-K rules.
- Explicit fallback decisions when simplification is unsafe or unnecessary.

Takeaway

The contribution is not just a model; it is a decision policy for interpretable longitudinal modeling.



04. Contributions

Summarize the contributions, extensions, and closing implications.

Contributions



1. Unified rectification-first pipeline

- Complete workflow from raw longitudinal data to sparse models and optional rules.

2. Scoped theoretical bridge

- Mechanistic explanation for why rectification can improve sparse recovery under explicit assumptions.

3. Anytime rule compression stage

- Converts active rectified support into compact m-of-K rule candidates when policy gates justify simplification.

4. Empirical validation package

- Synthetic, benchmark, HAI, UNICEF, ablation, and boundary studies characterize both gains and limits.

5. Reproducible implementation pathway

- CUTLASS package and notebooks connect the dissertation artifacts to reusable analyses.

Integrated Research Artifact

The dissertation contributes both a theoretical framing and a usable implementation pathway for interpretable longitudinal modeling.



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BACKUP SLIDES



RQ2: Putting the Pieces Together

Since:

$$|\theta_j| \leq \|C_{js}\|_{\infty} \|G_{ss}^{-1}\|_{\infty} \|\text{sign}(\beta_s)\|_{\infty}$$

and:

$$\|\tilde{C}_{js}\|_{\infty} \leq \|C_{js}\|_{\infty}, \quad \|\tilde{G}_{ss}^{-1}\|_{\infty} \leq \|G_{ss}^{-1}\|_{\infty}$$

we define:

$$B_j := \|C_{js}\|_{\infty} \|G_{ss}^{-1}\|_{\infty}, \quad \tilde{B}_j := \|\tilde{C}_{js}\|_{\infty} \|\tilde{G}_{ss}^{-1}\|_{\infty}$$

$$B^* := \max_{j \in S^c} (B_j), \quad \tilde{B}^* := \max_{j \in S^c} (\tilde{B}_j)$$

Thus, under the stated assumptions:

$$\Pr(\tilde{B}^* < 1) \geq \Pr(B^* < 1)$$

The lemmas isolate why binarization helps:

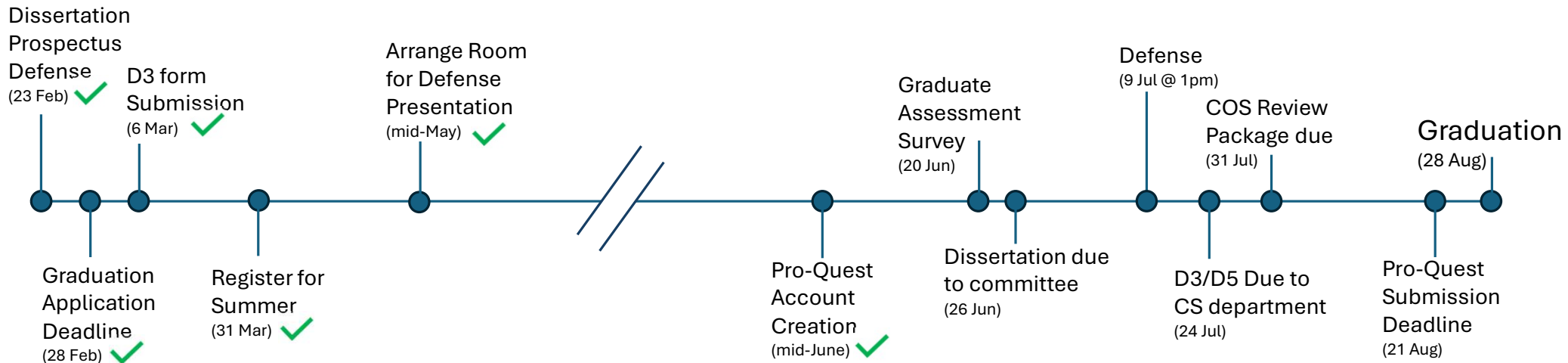
- (i) pairwise contraction keeps off-diagonals from saturating
- (ii) better conditioning bounds inverse norms
- (iii) Smaller off-support covariances reduce the IC term.

Together these raise the chance of correct support recovery and promote sparse, stable selections.

Scoped theoretical synthesis

Under the stated assumptions, sign binarization contracts pairwise dependence, improves inverse-Gram behavior, and reduces inactive-active leakage in the IC term. This supports the mechanism for improved support recovery, while practical rectification remains broader and requires empirical boundary checks.

Timeline To Graduation



Complete:

- All coursework
- D2 form (formation of committee) approved
- D9 form (advancement to candidacy) approved
- D3 form (prospectus approval) approved
- Graduation Application (online via LEO)
- Summer registration (completed)
- Room reservation (complete)